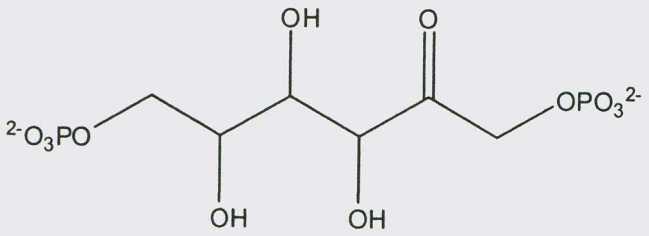
Biochemistry Cumulative Examination
October 1, 2010

Answer the following questions about glycolysis.

1. In the absence of oxygen pyruvate is commonly converted to lactate. Why? 10 points

2. The enzyme aldolase converts fructose 1,6-bisphosphate (shown below) into dihydroxyacetone phosphate and glyceraldehyde 3-phosphate. Show a mechanism for this reaction. 25 points



3. The enzyme triose phosphate isomerase converts dihydroxyacetone phosphate into glyceraldehyde 3-phosphate. Show a mechanism for this reaction. 25 points

4. 2-phosphoglycerate is converted into phosphoenolpyruvate by enolase. Both these molecules contain a phosphate attached to the small oxygen on the same carbon. One is considered a high energy phosphate bond and one is not. What is the difference that makes one high energy and explain briefly why it is higher energy. 15 points

5. There are several steps on glycolysis that are used to regulate the pathway. They catalyze reactions that are similar in what energetic aspect? 15 points

6. Some steps on glycolysis, such as the conversion of dihydroxyacetone phosphate into glyceraldehyde 3-phosphate are actually unfavorable. Why doesn't this prevent metabolic flux from moving through the pathway? 10 points